

Plan — R11 v2: Exact-Zero Generalized Hybrid Delivery + Hamiltonian Refinement

From the true zero reset $q = [0, 0, 0, 0]$ to a teacher-free, coin/target-conditioned hybrid manipulation policy, optimized for geometry, time, transition smoothness, and Hamiltonian energy flow — strict K6 and safety preserved.

HyMeKo coin-delivery campaign (recovery/coin-r9-causal-residual-delivery, parent 021a1e54)

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Abstract

R11.0 (**DONE**, 021a1e54) proved a positive control: from the exact zero reset a Cartesian reach + a READY-specific CEM capture + the frozen downstream delivers strict K6 (min_dtz 11.54 mm, coin untouched 0.0 mm, first frame $[0, 0, 0, 0]$). R11.1 (**DONE**) replaced the fragile 3-shell reach with a robust 4-DOF RRT-Connect that generalizes over the coin grid (9/10, ~ 0.5 s). The remaining bottleneck is *not* the reach but **capture/delivery generalization**: the learned R2 residual and the handoff orbit are canonical-only, so a coin shifted ≥ 2 –3 cm follows the wrong orbit and misses (28–66 mm). R11 (this plan) builds the rest: a HyMeKo hybrid IR (initial-condition certificate, modes $M_0 \rightarrow M_7$, transitions with handoff + energy descriptors), a coin/target demonstration bank, a **coin/target-conditioned teacher-free** capture–delivery policy (BC $\rightarrow$ reward-driven RL), and joint refinement of reach geometry, timing, hybrid-transition smoothness, and **Hamiltonian energy flow**. Every result starts from $q = [0, 0, 0, 0]$ in one continuous episode; strict K6, safety, no-premature-contact, and no-teleport are hard constraints. **This is a program-level plan; implementation is gated, one boundary at a time, R11.2 first.**

1 Scope and goal

Goal. A robust, coin/target-conditioned, *teacher-free* hybrid policy that generalizes from the exact zero home over $x_{\text{coin}} \sim \mathcal{D}_{\text{coin}}$, $x_{\text{target}} \sim \mathcal{D}_{\text{target}}$ and, subject to strict K6 + safety, minimizes path length, total $T_{\text{ZERO} \rightarrow \text{K6}}$, transition jerk/jumps, and Hamiltonian energy cost. **Deployed episode (one continuous rollout):**

EXACT_ZERO_HOME $\rightarrow$ collision-free reach $\rightarrow$ precontact alignment $\rightarrow$ capture $\rightarrow$ controlled delivery $\rightarrow$ target entry $\rightarrow$ settle $\rightarrow$ strict K6.

Non-goals / hard stops. No READY-snapshot start, no staging HOME, no un-executed pre-positioning, no transit omitted from the time/energy ledger, **no CEM/teacher at deployment**, and never report one planner’s failure as a physical impossibility. No RL run before its coordinate + reward + demo bank are frozen and its gate passes.

2 Frozen positive controls (R11.0 / R11.1 — do not modify)

- **R11.0 EXACT_ZERO_HOME_TO_STRICT_K6_BASELINE_PASS:** zero home $\rightarrow$ 3-shell IK reach $\rightarrow$ READY-specific CEM capture $\rightarrow$ frozen downstream $\rightarrow$ strict K6 (11.54 mm). CEM is a *teacher/oracle*, not deployment. The only learned component in this chain is the frozen R2 residual.

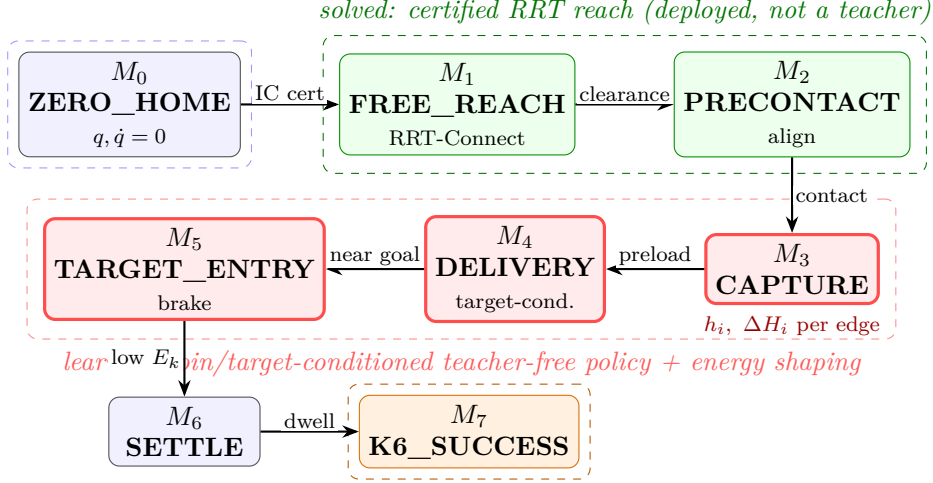


Figure 1: The hybrid mode automaton $M_0 \rightarrow M_7$. Each edge is a HybridTransition carrying a guard, a handoff descriptor h_i , and an energy descriptor ΔH_i . The reach ($M_0 \rightarrow M_2$) is a solved geometric planner (RRT-Connect); the learned, coin/target-conditioned policy owns capture through settle ($M_2 \rightarrow M_7$); energy-shaping targets ride on the transitions.

- **R11.1 ZERO_HOME_RRT_REACH_GRID_PASS_WITH_CLEARANCE_TIGHTENING_PENDING**: 4-DOF RRT-Connect, goal set of collision-free precontact configs, inflated-coin + inter-arm collision checks. Residual: $\sim 2\text{--}4\text{ mm}$ precontact nudge in a few executions (clearance tightening), and **right +5 cm** is an invalid initial condition.
- Frozen and reused (never the task start): the downstream (HANDOFF_RESET/R2/coast), physics, strict-K6 + safety monitors, the collision contract. **s4/s7** untouched; **f1--f4** sealed.
- **Two components, two roles — do not conflate. RRT-Connect is a certified, deterministic, deployed geometric planner** ($M_1 \rightarrow M_2$): it also serves as positive control, fallback, and an optional demonstration source, and is *legitimately present at final deployment* — the RL never has to relearn what the geometric planner solves robustly in $\sim 0.5\text{ s}$. Only the **per-instance CEM capture/delivery solve is teacher-only** (demonstration generator, prohibited at final deployment). *Teacher-free* therefore means: **no per-instance CEM / oracle / precomputed delivery orbit** — it does *not* mean “no deterministic RRT.”

3 Task contract (first-class, verifiable)

Initial condition EXACT_ZERO_HOME_V1: $q=[0, 0, 0, 0]$, $\dot{q}=0$, prev_tau= 0, empty controller memory, empty contact set, coin velocity 0, sim.step= 0, no snapshot parent, no teacher state. Coin/target poses are a *separate* init distribution, not part of HOME. **Every rollout** emits an INITIAL_CONDITION_CERTIFICATE (exact $q/\dot{q}$, zero memory, empty contacts, zero object velocity, step 0, no injection/staging); a ZERO_HOME $\rightarrow$ K6 claim is valid only with a passing certificate, over one continuous episode.

4 Hybrid system + energy model

Modes M_0 ZERO_HOME, M_1 FREE_REACH, M_2 PRECONTACT_ALIGNMENT, M_3 CAPTURE, M_4 CONTROLLED_DELIVERY, M_5 TARGET_ENTRY, M_6 SETTLE, M_7 K6_SUCCESS. **Transitions** carry $h_i = (q, \dot{q}, \tau_{\text{prev}}, x_c, \dot{x}_c, x_g, \mathcal{C}, e_{\text{phase}}, m_i, m_{i+1})$ and an EnergyTransitionCertificate. **Per-mode Hamiltonian** $H_m(z) = T_m + V_m + E_{\text{contact},m}$ with port-Hamiltonian dynamics $\dot{z} = (J_m(z) - R_m(z))\nabla H_m(z) + G_m(z)u$ ($J_m = -J_m^\top$ energy-conserving, $R_m \succeq 0$ dissipation). Reset $z^+ = \Delta_{m \rightarrow n}(z^-)$;

energy jump $\Delta H_{m \rightarrow n} = H_n(z^+) - H_m(z^-)$; episode ledger $H(T) - H(0) = W_{\text{act}} + W_{\text{ext}} - E_{\text{diss}} - E_{\text{impact}} + \varepsilon_{\text{num}}$. Positive and negative actuator work $W_{\pm} = \int \sum_i \max(\pm \tau_i \dot{q}_i, 0) dt$ are tracked *separately* (they must not cancel).

Two-level energy contract — do not claim conservation yet. Under MuJoCo contact, damping, actuator governance, and numerical integration, not every energy term is exactly recoverable. R11.2 delivers only the *measurement* level: a `MeasuredEnergyLedger` per mode/transition (robot E_k , coin E_k , PE where relevant, W_+ , W_- , contact impulse, damping/dissipation proxy, pre- and post-transition energy, numerical residual). The R11.2 `EnergyTransitionCertificate` asserts `ENERGY_LEDGER_COMPLETE` + `ENERGY_BALANCE_RESIDUAL_RECORDED` — never `HAMILTONIAN_CONSERVATION_PASS`. The *modelled* level (`ModelledHamiltonian`) which residual band is physically/numerically acceptable, and any conservation verdict) is calibrated later, at R11.8.

5 Coin/target-conditioned delivery (the real bottleneck)

Policy state $s_t = (q_t, \dot{q}_t, x_c, \dot{x}_c, x_g, \Delta x_{\text{goal}}, \mathcal{C}_t, h_t)$ with $\Delta x_{\text{goal}} = x_g - x_c$ (absolute pose kept — same relative vector needs different joint strategy at different workspace points). The policy *forms* the strategy from geometry; it does not store one canonical orbit. The **CEM teacher** re-derives the whole capture-delivery orbit per instance to build demonstrations (training only); **deployment uses no CEM, no oracle, no per-instance lookup, no precomputed delivery orbit** — but the deterministic RRT reach (M1→M2) *is* a legitimate deployed component (see §2). Teacher-free constrains the learned capture/delivery, not the certified geometric planner.

6 Distributions — certificate-filtered, not a raw rectangle

$\mathcal{D}_{\text{coin}} = (\text{geometric sample region}) \cap (\text{valid initial-condition set})$: a coin pose is *admissible* only if exact-zero HOME is collision-free, initial clearance ≥ 5 mm, coin inside environment bounds, a legal precontact goal set is nonempty, and RRT finds a certified reach within budget with precontact coin displacement ≤ 1 mm. **First region** about $c^* = [0.076, 0.143]$: $\Delta x \in [-0.05, +0.03]$, $\Delta y \in [-0.04, +0.03]$ m, *after* certificate filtering. **right +5 cm** is `INVALID_INITIAL_CONDITION` — excluded from the policy-success *denominator*, reported in a separate rejection-rate metric, never a “hard negative.” The current 2–4 mm precontact nudge must first be closed with larger obstacle inflation; only precontact-motion-gate-passing rollouts enter the demonstration bank. **Disjoint grids (no interpolation leakage, [[held-out panel is single-use]])**: *train* $\Delta x \in \{-4, -2, 0, +2\}$, $\Delta y \in \{-3, -1, +1, +3\}$ cm (admissible pairs); *dev* the half-shifted interior $\Delta x \in \{-3, -1, +1, +3\}$, $\Delta y \in \{-2, 0, +2\}$ cm (train-excluded); *final* edge-near admissible points, diagonals, continuous random cell-interior samples, plus a separate invalid-initial-condition rejection panel. **Target = relative goal vector, staged generalization.** **G0** translation control ($x'_c = x_c + \delta$, $x'_g = x_g + \delta$; relative delivery vector fixed — tests transfer to another absolute workspace point). **G1** target-direction ($\Delta x_g = r[\cos \alpha, \sin \alpha]$, $r \in [0.05, 0.09]$ m, $\alpha \in [\alpha^* - 45^\circ, \alpha^* + 45^\circ]$). **G2** full coin×direction×distance (only after G0/G1; each pair checked: target in bounds, no invalid overlap, a physical delivery path exists, the strict-K6 region fits). **R11.3 curriculum** C0 canonical/canonical → C1 shifted-together → C2 varied coin + fixed relative goal → C3 fixed coin + varied target dir/dist → C4 varied×varied → C5 unseen continuous panel — far more diagnostic than one large 2-D product distribution.

7 Optimization objective (lexicographic)

Hard: valid initial-condition certificate; collision/safety contract; no forbidden precontact coin motion; strict K6; teacher-free deployment. **Secondary** (over valid-K6 episodes): $J = \lambda_T T + \lambda_L L + \lambda_J J_{\text{jerk}} + \lambda_W W_+ + \lambda_D E_{\text{diss}} + \lambda_I E_{\text{impact}} + \lambda_H J_{\Delta H} + \lambda_S T_{\text{settle}}$, with $J_{\Delta H} = \sum_i \psi_i(\Delta H_i, W_{\text{act},i}, E_{\text{impact},i})$ and ψ_i

asymmetric (heavily penalize unexplained positive energy injection, penalize needless impact loss, permit physically-expected dissipation, reward hitting the mode-wise energy window). Constrained: $\min_{\pi} J$ s.t. $P(K6) \geq P(\text{baseline}) - \varepsilon$, $N_{\text{safety}} = N_{\text{invalid init}} = N_{\text{premature contact}} = 0$.

8 Affected files (new; no CORE.YAML items)

HyMeKo IR `hymeko_ir/` (`initial_condition`, `initial_distribution`, `initial_condition_certificate`, `hybrid_mode`, `hybrid_transition`, `transition_guard`, `state_invariant`, `handoff_descriptor`, `measured_energy_ledger`, `energy_transition_certificate`, `modelled_hamiltonian_certificate` [R11.8], `hamiltonian_mode` [R11.8], `rollout_provenance`). **Hybrid RL runtime** `hybrid_rl/` (`structured_plan`, `reach_planner_adapter`, `capture_delivery_policy`, `transition_policy`, `energy_shaping_policy`, `temporal_schedule`, `transition_blend`, `constrained_objective`, `trajectory_metrics`, `energy_ledger`, `positive_replay`, `deployment_guard`). **Coin adapter** `coin_delivery/` (`exact_zero_initial_condition`, `coin_target_distribution`, `rrt_reach` [from `coin_zero_home`, `precontact_goal_set`, `coin_collision_contract`, `capture_mode`, `delivery_mode`, `k6_certificate`, `coin_hamiltonian`, `coin_energy_metrics`, `coin_reward`). Reuse `option_rl` (SAC/TD3), `coin_zero_home_reach/rrt` (RRT is a *deployed* component), the frozen downstream + per-instance CEM as the *training* teacher only.

9 Gated phase ladder (each a stop-boundary; verdicts)

1. **R11.0 DONE** `EXACT_ZERO_HOME_TO_STRICT_K6_BASELINE_PASS`.
 2. **R11.1 DONE** `ZERO_HOME_RRT_REACH_GRID_PASS_WITH_CLEARANCE_TIGHTENING_PENDING`.
 3. **R11.2 (NEXT)** HyMeKo initial-condition + hybrid-mode/transition IR + certificates. Non-RL, verifiable. `INITIAL_CONDITION_CERTIFICATE_PASS` + IR round-trip tests. *This plan's first implementation boundary.*
 4. **R11.3** coin/target demonstration bank (sample $\rightarrow$ RRT reach $\rightarrow$ per-instance capture+delivery re-solve $\rightarrow$ K6/failure), disjoint train/dev/final grids. `COIN_TARGET_DEMONSTRATION_COVERAGE_PASS`.
 5. **R11.4** conditioned capture-delivery BC; teacher-free replay on part of the dev grid. `CONDITIONED_CAPTURE_DELIVERY_PASS`.
 6. **R11.5** reward-driven delivery generalization (SAC primary). `ZERO_HOME_COIN_TARGET_DELIVERY_GENERALIZATION_PASS`.
 7. **R11.6** reach geometry + timing refinement (contract frozen). `ZERO_HOME_REACH_REFINEMENT_PASS`.
 8. **R11.7** hybrid transition smoothing. `HYBRID_TRANSITION_SMOOTHING_PASS`.
 9. **R11.8** Hamiltonian energy refinement. `HAMILTONIAN_HYBRID_ENERGY_REFINEMENT_PASS`.
 10. **R11.9** joint geometry/timing/transition/capture/delivery/energy RL. `ZERO_HOME_GENERALIZED_HAMILTONIAN_HYBRID_PASS`.
 11. **R11.10** teacher-free final deployment (8+ seeds, unseen grid). `TEACHER_FREE_ZERO_HOME_GENERALIZED_DELIVERY_PASS`.
- See `plan.mmd` for the gated sequence.

10 R11.2 — first boundary (concrete, GO)

Deliverable (narrow scope): `InitialCondition`, `InitialDistribution`, `InitialConditionCertificate`, `HybridMode` ($M_0..M_7$), `HybridTransition`, `TransitionGuard`, `StateInvariant`, `HandoffDescriptor`, `MeasuredEnergyLedger`, `EnergyTransitionCertificate`, `RolloutProvenance`; the coin `exact_zero_initial_condition` adapter, the `k6_certificate` adapter, and RRT rollout instrumentation (emit IC certificate + provenance per rollout). All typed dataclasses with `debug_assert`-style validators (Design-by-Contract, §8). Purely additive; existing behavior unchanged. **Interfaces:** `InitialCondition.certify(rollout_state)  $\rightarrow$  Certificate|raises;` `HybridMode` enum + per-mode `StateInvariant`; `HybridTransition(source,target,guard,reset_map,handoff,energy_cert);` `RolloutProvenance` (git SHA, seed, IC cert, coin/target pose, mode/transition trace, measured ledger). **The 10 R11.2 gates:** `EXACT_ZERO_INITIAL_CONDITION_CERTIFICATE_PASS`, `INITIAL_DISTRIBUTION_VALIDITY_PASS`, `ROLLOUT_STATE_CONTINUITY_PASS`, `NO_SNAPSHOT_INJECTION_PASS`, `HYBRID_MODE_TRACE_VALID_PASS`, `HYBRID_TRANSITION_GUARD_VALID_PASS`, `HANDOFF_DESCRIPTOR_COMPLETE_PASS`, `MEASURED_ENERGY_LEDGER_COMPLETE_PASS`, `ENERGY_BALANCE_RESIDUAL_RECORDED_PASS`, `K6_CERTIFICATE_PROVENANCE_LINK_PASS`. The energy-ledger gate proves measurement *completeness*, not

optimality or conservation. **Tests:** certificate passes on a fresh zero-home rollout and *fails* on each violation (nonzero $\dot{q}$, nonempty contacts, injected snapshot, `step` $\neq$ 0, out-of-bounds coin); mode invariants; transition guard/reset round-trip; ledger balance recorded; provenance hash determinism. **Explicitly NOT in R11.2:** no demonstration generation, no CEM grid, no BC, no RL, no Hamiltonian reward, no energy shaping, no new planner, no policy training. Energy is *measured*, never optimized.

11 Test / performance / risk (§3, §4, program-level)

Tests at every boundary: unit (validators, invariants, metrics, energy ledger balance), integration (one continuous zero-home episode), and — for RL boundaries — a 1-seed production-scale smoke before any multi-seed queue. **Determinism:** fixed seeds; RRT seed pinned; energy ledger reproducible. **Perf:** demo-bank + RL run under the 16 GB RSS cap, checkpointed per seed; wall estimate reconciled ($> 2\times$ rule) before any overnight run. **Risk:** (i) delivery may not generalize even with conditioning (the R10.2/knife-edge precedent) — gated by `...GENERALIZATION_PASS`, else report `NO_GENERALIZATION_WITHIN_BUDGET`, never impossibility; (ii) energy-shaping could trade K6 for efficiency — the lexicographic hard-constraint on K6/safety forbids it; (iii) demo grids must be disjoint (train/dev/final) to avoid leakage ([[held-out panel is single-use]]). **Deployment guard:** deploy only if IC certificate valid, nominal K6 preserved, generalization $\geq$ baseline, zero safety regression, bounded precontact motion + energy injection, no unexplained transition energy jump, no CEM/teacher call, action in the certified envelope; else `DEPLOYMENT_FALLBACK_TO_RRT_PLUS_CERTIFIED_SCAFFOLD` (also from exact zero).

12 Rollback + hygiene

New `hymeko_ir/` + `hybrid_rl/` packages and coin-adapter files; rollback = delete. The IR wiring into the reach modules is additive (certificate + provenance emission), revertible without behavior change. Each RL checkpoint saved per seed; a regressing policy falls back to the RRT+certified-scaffold baseline. If R11 is abandoned before R11.2 lands, delete `docs/plans/2026-07-29-r11-generalized-hybrid-delivery/` this session.

13 Stop logic

Each numbered phase is a boundary: implement, verify its gate, write the report, and **HALT** for review before the next. No SAC/PPO/TD3 run before its boundary’s coordinate + reward + demo bank are frozen and its ranking/coverage gate passes. **This plan itself is the R11 boundary-0 artifact: no R11.2 code begins until it is approved.**